

Ajeet Krishnasamy

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Mechanical engineering graduate with hands-on experience building high-performance C++ physics solvers, including CFD and FEA tools for simulation, analysis, and structural design.

EXPERIENCE

Systems Engineer, Medical Equipment (Intern)

Oct 2020 – Sept 2021

CenTrak

Newtown, PA, United States

- Engineered automated reliability analysis pipelines for medical device telemetry, processing real-time performance data from 10,000+ RTLS tags to quantify failure rates and improve system monitoring precision.
- Developed automated performance dashboards for the Connect Pulse™ RTLS platform, enabling engineering teams to identify device degradation trends and reduce unplanned maintenance events.
- Conducted battery discharge and device health analysis across 5,000+ medical-grade IoT devices, identifying failure precursors and enabling predictive replacement scheduling that prevented equipment outages.
- Standardized structured query and analysis workflows for multi-terabyte RTLS datasets, reducing troubleshooting cycle time through automated anomaly detection and root-cause analysis scripts.

Systems Engineer, Medical Equipment (Intern)

July 2019 – Sept 2019

CenTrak

Newtown, PA, United States

- Analyzed RTLS location accuracy across hospitals, identified tracking blind spots from device data.
- Designed and executed validation test protocols for RTLS hardware deployment, correlating laboratory calibration data with field performance metrics to close the lab-to-field validation gap.
- Synthesized system performance data into technical recommendations for hardware upgrades and feature rollouts, supporting engineering decisions with quantitative evidence.
- Developed automated data analysis scripts for processing device telemetry, reducing manual analysis time and improving root-cause identification accuracy.

PROJECTS

AK-Vortex: High-Performance C++ Lattice Boltzmann CFD Suite | C++, Python, CFD

- Engineered a computational fluid dynamics (CFD) solver from scratch in C++ using the Lattice Boltzmann method with turbulence modeling and adaptive mesh refinement; validated against benchmark cases.
- Developed a desktop application in Rust with interactive geometry editing, real-time flow visualization, and contour plot generation.
- Implemented a machine-learning surrogate model that predicts flow fields, enabling rapid design-space exploration.

Crucible-FEA: 2D Finite Element Structural Solver | C++, Python, FEA

- Developed a 2D finite element analysis (FEA) solver in C++ supporting 6 element types for structural mechanics problems, with optimized sparse matrix solvers for efficient computation on large meshes.
- Implemented advanced capabilities including adaptive mesh refinement, nonlinear geometry handling, transient dynamics, and contact mechanics.
- Built a desktop application with a complete workflow: mesh generation, boundary condition assignment, real-time stress and displacement contour visualization.

Zenith: Orbital Mechanics & Flight Dynamics Engine | C++20, Python

- Built an orbital propagation engine in C++ implementing satellite orbit prediction (SGP4/SDP4) with 6 perturbation force models (J2, drag, solar radiation, third-body).
- Developed mission planning modules for conjunction assessment, orbital transfer maneuvers (Hohmann, Lambert solver), uncertainty propagation via Monte Carlo simulation, ground track prediction, and satellite constellation coverage analysis.
- Built a real-time 3D orbit visualization tool with interactive propagation controls, force model toggles, and a preset satellite database (ISS, Starlink, GPS, Galileo).

TECHNICAL SKILLS

Aerodynamics & CFD: ANSYS CFD, STAR-CCM+, SU2, OpenFOAM, ParaView

Programming & Simulation: C++, Python, MATLAB, OpenMP, PyTorch

CAD & Design: SolidWorks, Autodesk Fusion, GD&T, Finite Element Analysis (FEA)

Manufacturing: CNC Machining, 3D Printing, Lathe & Mill Operations, MIG Welding

EDUCATION

University of Ottawa

Ottawa, ON, Canada

Bachelor of Applied Science (BASc), Biomedical Mechanical Engineering

May 2026

Coursework: Fluid Mechanics, Heat Transfer, Control Systems, Vibration Analysis, Machine Design, Dynamics

Accreditation: CEAB-accredited; ABET-equivalent